

Automatic Musical Instrument Recognition with MERT

From Isolated MedleyDB Stems to Polyphonic Mixtures

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Abstract

Automatic musical instrument recognition is a central task in Music Information Retrieval, with applications in music indexing, source-aware analysis, recommendation, and production-oriented tools. This work investigates instrument classification on MedleyDB v1, a royalty-free multitrack dataset that provides isolated stems and metadata for real music recordings. We formulate the problem in two settings: single-label classification of isolated instrument stems and multi-label recognition in controlled polyphonic mixtures. A classical baseline based on MFCC and spectral descriptors with an RBF-SVM is compared with transfer-learning systems based on MERT, a self-supervised music audio representation model. The experimental pipeline includes track-disjoint dataset construction, five-second segmentation, class balancing, frozen-embedding classifiers, MERT layer and model-size ablations, partial fine-tuning, protocol variants, and synthetic mixture generation with one to four active instruments. On the balanced 12-class isolated-stem task, the MFCC/SVM baseline reaches a test macro-F1 of 0.6752, while the initial frozen MERT-v1-95M reference reaches 0.6431. Representation choice is crucial: intermediate-layer and learned-layer MERT embeddings improve performance, and MERT-v1-330M with learned all-layer aggregation obtains the highest observed isolated-stem test macro-F1 of 0.7290. Last-layer fine-tuning of MERT-v1-95M gives only a small test improvement and no validation-based advantage. In the polyphonic setting, the frozen MERT-v1-95M multi-label classifier obtains a macro-F1 of 0.5353 and a macro ROC-AUC of 0.7867, but exact-match accuracy remains low and decreases as the number of overlapping instruments increases. Overall, the results show that self-supervised music representations are effective for instrument recognition when the appropriate representation layer is selected, while robust complete-set prediction in polyphonic mixtures remains substantially more challenging.

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1 Introduction

Automatic musical instrument recognition is a core task in Music Information Retrieval (MIR), with applications in music indexing, automatic annotation, recommendation, transcription support, and audio production tools. The goal is to identify which instrument is active in an audio recording, either from isolated stems or from mixtures containing multiple simultaneous sources. This task is challenging because musical instruments exhibit large timbral variability depending on pitch range, playing technique, performer, recording conditions, and production effects. The problem becomes even more difficult in polyphonic music, where several instruments overlap and partially mask each other.

This project addresses automatic instrument classification using MedleyDB v1, a royalty-free multitrack dataset containing professionally recorded songs, isolated stems, metadata, and instrument annotations [1]. The availability of individual stems makes MedleyDB suitable for studying both isolated-stem classification and controlled polyphonic recognition. In this work, the isolated-stem setting is formulated as a 12-class single-label classification problem, while the polyphonic setting is formulated as a multi-label task using synthetic mixtures with one to four active instruments.

The proposed baseline follows a classical audio-classification pipeline based on handcrafted timbral descriptors. In particular, MFCC and spectral features are extracted from five-second audio segments and classified using an RBF-kernel Support Vector Machine. This baseline is simple, computationally efficient, and directly related to traditional MIR approaches based on feature engineering. As a state-of-the-art alternative, we evaluate transfer learning with MERT, a self-supervised music audio representation model designed to learn general-purpose acoustic representations for MIR tasks [2]. MERT is selected because it is music-specific, pretrained on large-scale audio data, and can be used as a frozen feature extractor or partially fine-tuned for downstream classification.

The experimental pipeline includes audio preprocessing, stem selection, five-second segmentation, song-level train/validation/test splitting, feature extraction, model training, and objective evaluation. To avoid data leakage, segments from the same song are not shared across partitions. Model performance is assessed using accuracy and macro-F1 for isolated-stem classification, and macro-F1, micro-F1, ROC-AUC, and exact-match accuracy for polyphonic multi-label classification. Additional experiments investigate MERT model size, representation layer, pooling strategy, classifier head design, and partial fine-tuning.

The results show that handcrafted features provide a strong baseline, but that the choice of learned representation is crucial. The MFCC/spectral SVM baseline reaches a test macro-F1 of 0.6752 on the isolated-stem task, outperforming the initial frozen MERT-v1-95M final-layer reference. However, improved MERT configurations, especially intermediate or learned layer aggregation and the larger MERT-v1-330M model, achieve higher performance, with the best isolated-stem configuration reaching a test macro-F1 of 0.7290. In the polyphonic setting, the MERT-based multi-label classifier obtains informative label-wise predictions, but exact complete-set recognition remains difficult and degrades as the number of overlapping instruments increases.

Overall, this work provides a complete and reproducible comparison between a classical feature-engineering baseline and self-supervised music representations for MedleyDB instrument classification. The study shows that pretrained music models can improve isolated-stem recognition when the representation is carefully selected, while robust recognition in dense polyphonic mixtures remains an open challenge.

2 Background and Related Work

Traditional musical instrument recognition systems often rely on handcrafted timbral descriptors such as MFCCs, spectral centroid, spectral bandwidth, roll-off, zero-crossing rate, and energy statistics. These features are usually combined with classical classifiers such as SVMs, k-NN, or random forests. This approach is efficient and interpretable, but it may miss longer-term musical context and more complex spectro-temporal patterns that are relevant when instruments appear with different playing styles, registers, and production effects.

Recent approaches increasingly use pre-trained audio models to extract more general representations from waveform audio. MERT is a music-specific self-supervised model designed for acoustic music understanding [2]. Its pre-training follows a masked-prediction strategy: parts of the audio representation are masked, and the model learns to recover pseudo-labels produced by teacher models. In particular, the original paper uses both an acoustic teacher and a musical teacher based on CQT information, so that the learned representation captures not only general audio structure but also pitch- and harmony-related musical information.

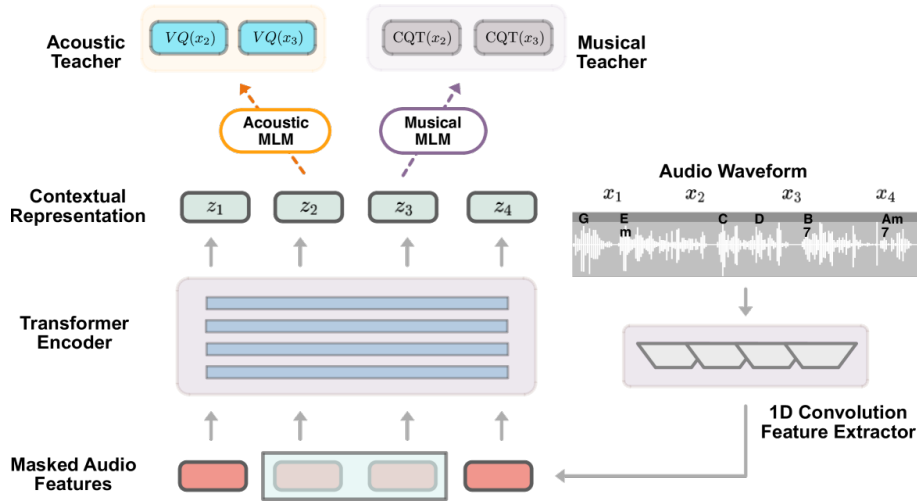


Figure 1: Overview of the MERT self-supervised pre-training framework, adapted from [2].

This makes MERT a suitable state-of-the-art representation model for this project. Instead of training a deep audio model from scratch on the relatively small MedleyDB subset, MERT can be used as a frozen feature extractor and combined with lightweight classifiers. Recent studies have also shown the potential of self-supervised music models for multitrack and polyphonic instrument analysis [3, 4], supporting the later extension of this work from isolated stems to controlled instrument mixtures.

3 Dataset and Data Preparation

3.1 MedleyDB v1

This project uses the original release of MedleyDB, here referred to as MedleyDB v1. MedleyDB is a royalty-free multitrack dataset designed for annotation-intensive Music Information Retrieval research [1, 5]. The original release contains 122 songs, while the later MedleyDB 2.0 extension is not used in this work.

Each MedleyDB song includes a final mix, processed stems, raw audio files, metadata, and annotations. The processed stems are particularly suitable for instrument recognition because they correspond to musically meaningful sources with associated instrument labels and are temporally

aligned with the full song. The dataset was originally introduced for melody extraction, but its multitrack structure also makes it useful for instrument identification, source separation, and automatic mixing.

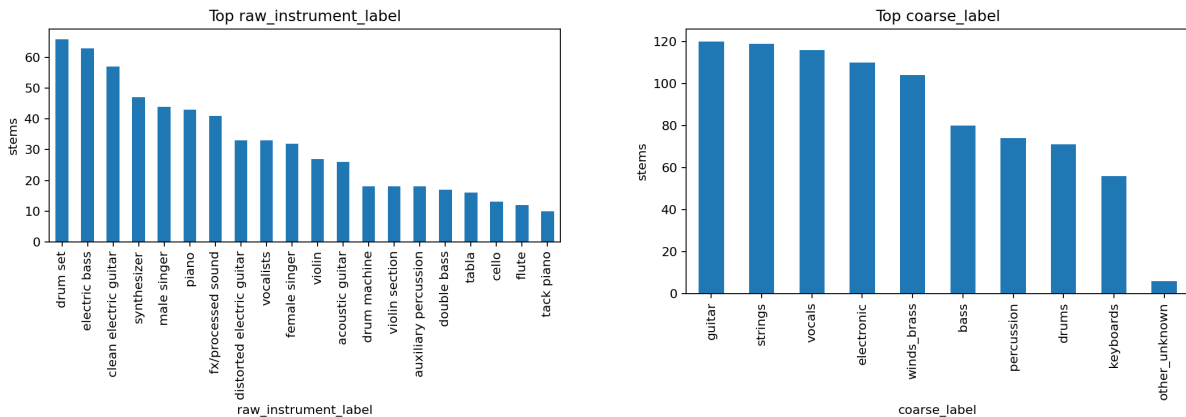
The audio is distributed as WAV files at 44.1 kHz and 16-bit resolution. Mixes and stems are stereo, whereas raw recordings are mono. In this project, the isolated-stem experiments use the processed stems rather than the raw recordings, because the goal is to recognise instruments as they appear in a real music-production context. Stems may therefore include production choices such as panning, equalisation, compression, or reverb.

MedleyDB includes music from several genres and recording sources. Most songs are full-length recordings, and the dataset contains both vocal and instrumental material. Some recordings were captured in live or semi-live conditions, so certain stems may contain microphone bleed from other instruments. For the isolated-stem experiments, stems marked as containing bleed are excluded in order to keep the single-label task as clean as possible. Source overlap is then studied separately in the polyphonic-mixture experiments.

3.2 Labels and subset construction

The experiments use MedleyDB instrument-level labels, but broad instrument families labels were also used during dataset exploration. The labels are only lightly standardised for formatting and spelling, so acoustically distinct categories such as clean electric guitar and distorted electric guitar remain separate classes.

The full MedleyDB label distribution is highly imbalanced: common instruments such as drums, bass, piano, vocals, and guitars appear much more frequently than rare instruments, as shown in Figure 2. To obtain a reliable and interpretable classification task, the main isolated-stem experiments use a balanced subset built from the best represented instrument classes. A class is retained only when it has enough segments and enough different source tracks to support training, validation, and testing.



(a) Top raw instrument labels.

(b) Coarse instrument families.

Figure 2: Instrument-label distributions before subset balancing.

Each selected stem is divided into non-overlapping 5 s excerpts. This duration is a compromise between musical context and dataset size. Shorter windows may miss relevant timbral cues such as attacks, sustain, articulation, or short musical phrases, while longer windows would reduce the number of available examples and increase the computational cost of MERT feature extraction. Very low-energy excerpts are removed before training so that mostly silent windows, below 50 dBFS, do not become misleading examples.

The final isolated-stem subset contains 12 instrument classes and uses equal per-class quotas. This avoids domination by the most frequent instruments and makes macro-averaged metrics meaningful. The resulting split contains 1344 training segments, 288 validation segments, and 288 test segments, with all segments from the same song kept in the same partition.

Table 1: Default isolated-stem subset configuration.

Parameter	Value
Dataset release	MedleyDB v1
Audio source	Processed stems
Label type	MedleyDB instrument-level labels
Segment duration	5 s
Hop duration	5 s
Number of classes	12
Segments per class	160
Train/validation/test ratio	70/15/15
Bleed policy	Stems marked with bleed excluded

3.3 Track-disjoint splitting

Splits are created at the song level, so all stems and segments from the same recording remain in one partition. The final dataset contains 60 training, 13 validation, and 13 test tracks, with no overlap between splits. This avoids leakage from shared performers, recording conditions, production processing, and musical context.

3.4 Waveform preprocessing

Segments are converted to mono, resampled, and safely peak-normalised to 0.99. The classical baseline uses 22.05 kHz, while MERT uses 24 kHz. Activity filtering is performed before normalisation to avoid retaining artificially amplified silent regions.

4 System Methodology

4.1 Experimental pipeline

The project follows a configuration-driven pipeline covering dataset preparation, representation extraction, classifier training, evaluation, and experiment tracking. MedleyDB songs are divided into track-disjoint training, validation, and test partitions before segment extraction, ensuring that all segments from the same song remain in the same split.

Each experiment stores its resolved configuration, predictions, aggregate and per-class metrics, confusion matrices, and checkpoints. Frozen MERT representations are additionally cached to avoid repeating encoder inference.

4.2 Isolated-stem classification

The isolated-stem task is formulated as 12-class single-label instrument classification. A classical baseline uses temporally aggregated MFCC and spectral descriptors followed by an RBF-kernel SVM.

The neural baseline uses frozen MERT embeddings. Audio is converted to mono, resampled to 24 kHz, peak-normalised, and processed by the MERT encoder. The operational reference

configuration in the main pipeline uses MERT-v1-95M, the final Transformer layer, temporal mean pooling, and a one-hidden-layer MLP with 256 hidden units and dropout 0.2.

The first experimental stage compares alternative classifier heads, including a linear classifier and MLP variants with different hidden dimensions, dropout values, and learning rates. The best validation-selected frozen-MERT head, an MLP with 256 hidden units and dropout 0.5, is then kept fixed during the representation ablations. This separates the operational reference run from the classifier-head setting used for the later representation ablations.

4.3 MERT representation ablations

The representation experiments compare different Transformer layers, averages of multiple layers, learned layer aggregation, temporal pooling strategies, and the MERT-v1-95M and MERT-v1-330M checkpoints. These experiments use cached embeddings and the same downstream classifier so that performance differences can be attributed mainly to the selected representation.

All single-label neural classifiers are trained with AdamW and cross-entropy loss. Validation macro-F1 is used for early stopping and model selection, while the test partition is evaluated only after the final checkpoint has been selected.

4.4 Direct-audio training and partial fine-tuning

The direct-audio experiments process the waveform through MERT during every training iteration. They compare a fully frozen MERT-v1-95M encoder with fine-tuning of the final Transformer layer.

The classifier head uses a larger learning rate than the unfrozen encoder parameters. Mixed precision, gradient accumulation, and gradient checkpointing are used to reduce GPU-memory requirements. Full-backbone fine-tuning is not considered because of the available computational resources and the limited dataset size.

4.5 Protocol ablations

The protocol experiments keep the frozen MERT-v1-95M final-layer mean-pooling representation fixed and use the selected MLP head with 256 hidden units and dropout 0.5. They modify only the data or training setup. The completed comparisons investigate different segment durations, a reduced balanced subset, and a capped-natural subset that preserves unequal class frequencies while guaranteeing minimum class coverage.

These experiments evaluate whether the observed performance depends on segment length, dataset size, or class distribution rather than only on the model architecture.

4.6 Polyphonic dataset construction

The final stage extends the isolated-stem pipeline to multi-label instrument recognition. The reported experiments use a protocol in which between one and four isolated instrument segments are sampled from the same dataset partition and combined into a synthetic mixture.

Each source is peak-normalised, the selected signals are averaged, and the resulting mixture is normalised again. The generator stores a deterministic manifest containing source paths, offsets, labels, overlap level, split, and random seed. Mixtures are reconstructed from this manifest when needed rather than being permanently rendered to disk.

The target of each mixture is a 12-dimensional multi-hot vector indicating all active instruments.

4.7 Multi-label classification

The completed polyphonic experiments use frozen MERT-v1-95M embeddings from the final layer with temporal mean pooling, matching the operational reference representation from the main pipeline. The classifier is a one-hidden-layer MLP with 256 hidden units, ReLU activation, dropout 0.2, and 12 independent output logits.

Training uses weighted binary cross-entropy with logits, where class weights are computed from the training-set positive and negative frequencies. Validation macro-F1 is used for early stopping and checkpoint selection. During evaluation, sigmoid probabilities are converted into binary predictions using a fixed threshold of 0.5.

5 Experimental Programme

The experimental programme follows the implementation order reported in Section 7:

1. **Reference pipeline.** A complete frozen-MERT pipeline is executed on the balanced isolated-stem dataset to verify the workflow and establish the operational reference configuration: MERT-v1-95M, final-layer mean pooling, and an MLP with 256 hidden units and dropout 0.2.
2. **Classical baseline and classifier heads.** An MFCC and spectral-feature SVM is compared with linear and MLP classifiers trained on frozen MERT-v1-95M embeddings.
3. **MERT representation ablations.** Different Transformer layers, layer-aggregation methods, temporal-pooling strategies, and the MERT-v1-95M and MERT-v1-330M checkpoints are compared using a fixed downstream classifier.
4. **Direct-audio training.** A directly evaluated frozen MERT model is compared with partial fine-tuning of the final Transformer layer.
5. **Protocol ablations.** The frozen MERT-v1-95M final-layer system with the selected dropout-0.5 MLP head is evaluated with different segment durations, a reduced balanced subset, and a capped-natural class distribution with plain training.
6. **Polyphonic dataset and classifier.** Synthetic mixtures containing between one and four active instruments are generated from the isolated-stem partitions. Frozen MERT embeddings are then used to train a 12-class multi-label classifier.
7. **Polyphonic analysis.** The trained model is evaluated as a function of the number of active instruments. Instrument frequency, genre relationships, synthetic co-occurrence, recurrent errors, and representative predictions are also analysed.

Validation macro-F1 is used for neural-model selection, while the held-out test partition is evaluated only after the final configuration has been selected. The completed polyphonic experiments retain the operational MERT-v1-95M final-layer mean-pooling configuration with the dropout-0.2 MLP, rather than the highest-scoring isolated-stem representation or the dropout-0.5 head selected for some isolated-stem ablations.

6 Evaluation Protocol

6.1 Single-label evaluation

For C instrument classes, precision, recall, and F1-score are computed for each class. Macro-F1 is defined as

$$F_{1,\text{macro}} = \frac{1}{C} \sum_{c=1}^C F_{1,c}, \quad (1)$$

and is the primary metric because it assigns equal importance to every class, regardless of its support. Accuracy and weighted-F1 are reported as complementary measures. Confusion matrices, per-class metrics, and prediction-level outputs are retained for further analysis.

6.2 Model selection

Neural models are selected using validation macro-F1, which is also monitored for early stopping. The checkpoint with the highest validation score is then evaluated once on the held-out test set. The test partition is never used for training, hyperparameter selection, or checkpoint selection.

The fixed classical baseline does not use validation-based model selection. As described in Section 4.2, its final classifier is trained on the combined training and validation partitions and evaluated on the same held-out test set.

6.3 Polyphonic evaluation

The multi-label extension is evaluated using exact-match accuracy, micro-precision, micro-recall, micro-F1, macro-precision, macro-recall, macro-F1, mean average precision, macro ROC-AUC, and per-class scores. Exact-match accuracy counts a prediction as correct only when the complete predicted label set is identical to the reference label set. Results are also grouped by the number of active instruments, k_{active} , to measure the effect of increasing source overlap.

A fixed threshold of 0.5 is applied to every sigmoid output. Validation macro-F1 is used for checkpoint selection, but threshold tuning is not performed in the reported run.

6.4 Error analysis

Quantitative results are complemented by analysis of frequent false positives, false negatives, weak classes, high-confidence exact-match errors, and representative correct predictions. These errors are interpreted in relation to class support, timbral similarity, genre context, instrument co-occurrence, and the controlled synthetic-mixture construction. The analysis uses saved prediction outputs and representative mixture-level examples rather than a systematic microphone-bleed study.

7 Results

This section reports the experiments in the order in which they are implemented in the project notebooks. The first complete pipeline execution establishes the reference configuration. The following notebooks then investigate classifier capacity, MERT representations, partial fine-tuning, data robustness, and the extension to polyphonic mixtures.

Main Pipeline: Reference Run

The first complete execution of `pipeline_medleydb.ipynb` verifies the end-to-end workflow and establishes the frozen-MERT reference system, summarised in Table 2, used in the subsequent comparisons. Since the data-processing and training pipeline have already been described in the previous sections, only the resulting dataset, model configuration, and evaluation results are reported here.

Table 2: Dataset and model configuration of the reference run.

Component	Configuration
Subset	largest_balanced, medleydb_instrument
Segments	5 s duration, 5 s hop, 160 segments per class
Data split	1344 training, 288 validation, and 288 test segments
Track separation	Track-disjoint, with 13 distinct tracks in the test set
Encoder	Frozen MERT-v1-95M
Representation	Final hidden layer with temporal mean pooling, 768 dimensions
Classifier	MLP with 256 hidden units and dropout 0.2

The retained instrument classes are listed in Table 4. Each class contributes 112 training, 24 validation, and 24 test segments.

Overall performance

The best checkpoint was obtained at epoch 21, with a validation macro-F1 of 0.6146. On the held-out test set, the model correctly classified 186 of 288 segments.

Table 3: Validation and held-out test performance of the reference run.

Metric	Value
Best validation macro-F1	0.6146
Best validation accuracy	0.6563
Test accuracy	0.6458
Test macro-F1	0.6431
Correct test predictions	186 / 288

Because every test class has the same support, test macro-F1 and weighted-F1 are both 0.6431.

Per-class performance

Table 4: Per-class performance of the reference model. Each class has a test support of 24 segments.

Class	Precision	Recall	F1
Acoustic guitar	0.960	1.000	0.980
Auxiliary percussion	0.857	0.500	0.632
Clean electric guitar	0.448	0.542	0.491
Distorted electric guitar	0.500	0.292	0.368
Drum set	0.960	1.000	0.980
Electric bass	0.870	0.833	0.851
Female singer	0.688	0.458	0.550
FX/processed sound	0.417	0.208	0.278
Male singer	0.636	0.875	0.737
Piano	0.870	0.833	0.851
Synthesizer	0.271	0.542	0.361
Vocalists	0.615	0.667	0.640

The strongest results are obtained for *acoustic guitar* and *drum set*, both with F1 0.980, followed by *electric bass* and *piano*, both with F1 0.851. The weakest classes are *FX/processed sound*, *synthesizer*, and *distorted electric guitar*.

Although acoustic guitar achieves perfect recall, its test segments originate from only one track. This result should therefore be interpreted cautiously: strong segment-level performance does not necessarily demonstrate generalisation across different recordings.

Confusion-matrix analysis

The confusion matrices reveal three main areas of ambiguity. First, 11 of the 24 *auxiliary percussion* segments are classified as *synthesizer*. Second, distorted electric guitar is frequently confused with both clean electric guitar and synthesizer: only seven segments are classified correctly, while eight are assigned to clean electric guitar and nine to synthesizer.

The vocal categories also partially overlap. Seven female-singer segments are classified as male singer and four as vocalists. Although male singer reaches a recall of 0.875, its lower precision indicates that the class also receives predictions belonging to the other vocal categories.

The synthesizer label is assigned to 48 test segments despite having a true support of 24. This over-prediction explains its combination of moderate recall and low precision. The *FX/processed sound* category produces errors across several classes and obtains the lowest F1, consistently with its heterogeneous acoustic content.

The raw and row-normalised confusion matrices are shown side by side in Figure 3. The raw matrix reports the number of segments associated with each prediction, whereas the normalised matrix shows the prediction distribution within each true class.

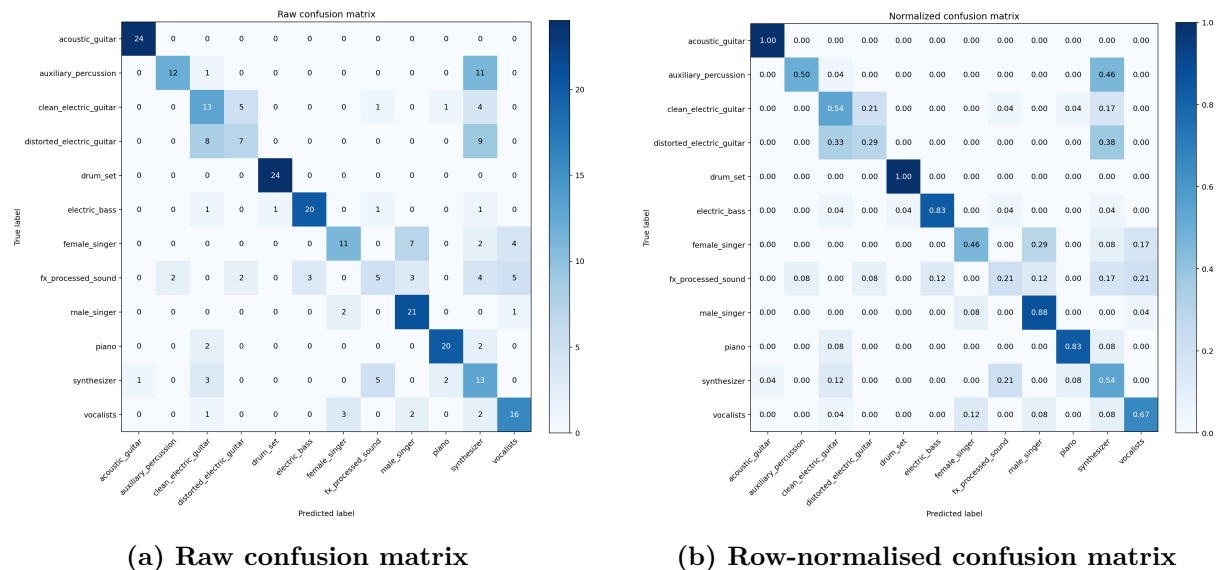


Figure 3: Confusion matrices of the main frozen-MERT reference pipeline. The raw matrix reports the number of test segments, while the row-normalised matrix reports the prediction proportions for each true class.

7.1 Notebook 1: Classical Baseline and Classifier-Head Comparison

Notebook 1 compares the classical MFCC-based baseline with different classifier heads trained on frozen MERT-v1-95M embeddings. All experiments use the same balanced 12-class stem subset with MedleyDB instrument-level labels described in 1.

The frozen-MERT experiments use the final hidden layer with temporal mean pooling. The comparison includes a linear classifier, MLPs with different dropout values and hidden

dimensions, and a lower-learning-rate variant. Since the input embeddings and data partitions remain unchanged, differences between these runs can be attributed to the classifier-head and optimisation settings.

7.1.1 Overall comparison

The results are summarised in Table 5. Weighted-F1 is not reported separately because the balanced test set gives every class the same support, making it identical to macro-F1.

Table 5: Classical baseline and frozen-MERT classifier-head comparison.

Configuration	Val. macro-F1	Test accuracy	Test macro-F1
MFCC/spectral SVM	–	0.6806	0.6752
MERT + linear head	0.5946	0.6424	0.6377
MERT + MLP 256, dropout 0.0	0.6062	0.6319	0.6188
MERT + MLP 256, dropout 0.2	0.6146	0.6458	0.6431
MERT + MLP 256, dropout 0.5	0.6296	0.6563	0.6497
MERT + MLP 512, dropout 0.2	0.6000	0.6389	0.6364
MERT + MLP 256, dropout 0.2, LR 3×10^{-4}	0.6043	0.6563	0.6438

The MFCC/spectral-descriptor SVM obtains the highest held-out test performance, with an accuracy of 0.6806 and a macro-F1 of 0.6752, corresponding to 196 correctly classified segments out of 288. The best frozen-MERT model is the MLP with 256 hidden units and dropout 0.5, which reaches a validation macro-F1 of 0.6296, a test accuracy of 0.6563, and a test macro-F1 of 0.6497. It correctly classifies 189 test segments.

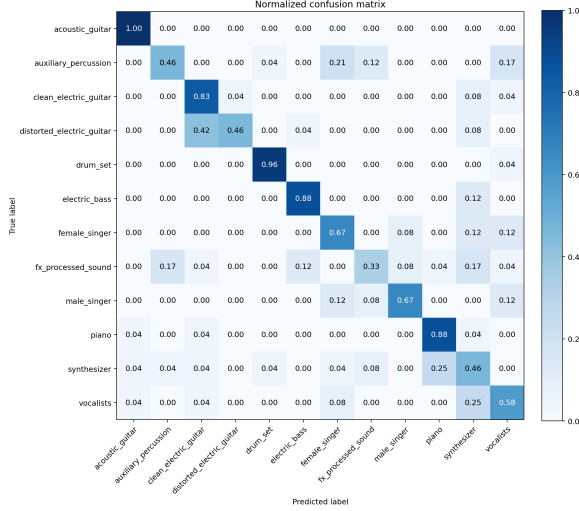
The classical baseline therefore exceeds the best frozen-MERT head by approximately 2.4 percentage points in accuracy and 2.6 percentage points in macro-F1. However, the SVM is a fixed reference model trained on the combined training and validation partitions and does not have a validation-selection score. Its higher test result should therefore be interpreted as a descriptive comparison rather than as validation-based model selection.

Among the frozen-MERT configurations, dropout improves performance. Increasing dropout from 0.0 to 0.5 raises validation macro-F1 from 0.6062 to 0.6296 and test macro-F1 from 0.6188 to 0.6497. In contrast, increasing the hidden dimension from 256 to 512 does not provide an advantage, suggesting that additional classifier capacity is unnecessary for the fixed 768-dimensional representation.

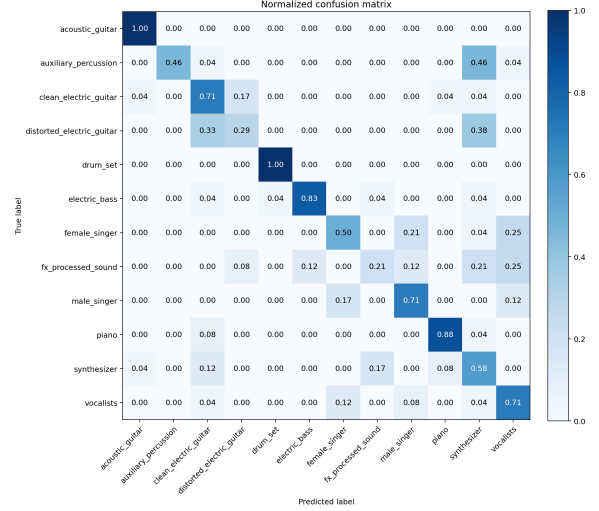
Reducing the learning rate from 10^{-3} to 3×10^{-4} produces the same test accuracy as the dropout-0.5 model, but a lower validation macro-F1 and a lower test macro-F1. The linear classifier is competitive with several MLP variants, but remains below the best regularised MLP. Overall, the results indicate that moderate classifier capacity combined with stronger dropout is the most effective head configuration for the frozen last-layer embeddings.

7.1.2 Error analysis

To avoid reporting a separate confusion matrix for every classifier, only the row-normalised matrices of the highest-scoring classical model and the best frozen-MERT model are compared in Figure 4.



(a) MFCC/spectral SVM



(b) Frozen MERT, MLP 256, dropout 0.5

Figure 4: Row-normalised confusion matrices for the best classical baseline and the best frozen-MERT classifier head in Notebook 1.

Both models classify acoustic guitar and drum set reliably, but they exhibit different error patterns. The most frequent SVM error is the classification of 10 distorted-electric-guitar segments as clean electric guitar. Nevertheless, the SVM obtains substantially higher F1 scores for both clean and distorted electric guitar than the best MERT model, particularly for distorted guitar (0.611 against 0.378).

The frozen-MERT model shows a stronger tendency to assign ambiguous examples to the synthesizer class. Eleven auxiliary-percussion segments are predicted as synthesizer, while distorted electric guitar is confused with synthesizer in nine cases and with clean electric guitar in eight cases. Vocal and processed sound categories also overlap: female singer is frequently assigned to vocalists or male singer, and several FX/processed-sound segments are predicted as vocalists or synthesizer.

Despite its lower aggregate score, the best frozen-MERT model improves over the SVM for some classes, including drum set, piano, auxiliary percussion, synthesizer, and vocalists. The SVM advantage is concentrated mainly in the electric-guitar classes, FX/processed sound, and the individual female- and male-singer categories.

These results show that the classifier head alone does not allow the frozen last-layer MERT representation to surpass the handcrafted baseline. The dropout-0.5 MLP is therefore retained as the strongest classifier head for the frozen-MERT system and is used as the reference head in the representation ablations of Notebook 2.

7.2 Notebook 2: MERT Representation Ablations

Notebook 2 investigates how the choice of MERT representation affects instrument-classification performance. All experiments use the same dataset and track-disjoint partitions adopted in the previous experiments.

To isolate the effect of the representation, the classifier is kept fixed as an MLP with 256 hidden units and dropout 0.5. The experiments compare intermediate and final Transformer layers, averages of multiple hidden states, learned all-layer aggregation, different temporal-pooling methods, and the MERT-v1-95M and MERT-v1-330M checkpoints.

7.2.1 Representation comparison

The complete comparison is reported in Table 6. The highest validation and held-out test scores are highlighted separately because they are obtained by different configurations.

Table 6: MERT representation ablation results. All configurations use the same MLP classifier with 256 hidden units and dropout 0.5.

Representation	Val. macro-F1	Test accuracy	Test macro-F1
MERT-95M, layer 6, mean pooling	0.7025	0.7118	0.7007
MERT-95M, layer 9, mean pooling	0.6384	0.6632	0.6512
MERT-95M, final layer, mean pooling	0.6146	0.6458	0.6431
MERT-95M, average of final 3 layers, mean pooling	0.6284	0.6632	0.6532
MERT-95M, average of final 6 layers, mean pooling	0.6326	0.6910	0.6783
MERT-95M, learned all-layer aggregation, mean pooling	0.6495	0.7257	0.7117
MERT-95M, final layer, max pooling	0.5372	0.5938	0.5681
MERT-95M, final layer, mean-max pooling	0.5664	0.6667	0.6542
MERT-330M, final layer, mean pooling	0.7337	0.7222	0.7091
MERT-330M, learned all-layer aggregation, mean pooling	0.7002	0.7500	0.7290

The results show that the final layer is not the most effective source of instrument information. Within MERT-v1-95M, layer 6 reaches a validation macro-F1 of 0.7025 and a test macro-F1 of 0.7007, improving the final-layer reference by 5.8 percentage points on the test set. This suggests that intermediate representations preserve timbral information that becomes less directly accessible in the final Transformer layers.

Averaging the final six layers also improves over the final-layer reference, but remains below layer 6. Learned aggregation of all MERT-v1-95M hidden states produces the highest test macro-F1 among the 95M configurations (0.7117), although its validation score is lower than that of layer 6. The difference between validation and test rankings indicates that small score variations should not be interpreted as definitive evidence that one aggregation method is universally superior.

Temporal max pooling performs worst, with a test macro-F1 of 0.5681. Concatenating mean and max pooling improves this result but does not outperform mean pooling alone. These results indicate that temporal averaging provides a more stable summary of the five-second segments than selecting only the largest activation at each feature dimension.

7.2.2 Effect of model size

Both MERT-v1-330M configurations outperform the original MERT-v1-95M final-layer reference. With final-layer mean pooling, increasing the encoder size raises test macro-F1 from 0.6431 to 0.7091. Learned all-layer aggregation with MERT-v1-330M obtains the highest observed held-out test result, with an accuracy of 0.7500 and a macro-F1 of 0.7290. The model correctly classifies 216 of the 288 test segments, 30 more than the original frozen-MERT reference run.

The predefined evaluation protocol selects configurations using validation macro-F1 rather than test performance. By this criterion, the MERT-v1-330M final-layer mean representation is the selected configuration, with validation macro-F1 0.7337 and test macro-F1 0.7091. The all-layer MERT-v1-330M result should therefore be described as the highest observed test score, rather than selected solely on the basis of the held-out test set.

Nevertheless, both configurations provide clear evidence that the larger MERT checkpoint improves over the original MERT-v1-95M final-layer representation. The learned all-layer

configuration also exceeds the MFCC/SVM baseline from Notebook 1, whose test macro-F1 was 0.6752.

7.2.3 Error analysis of the highest-scoring test configuration

The row-normalised confusion matrix of the MERT-v1-330M learned all-layer model is shown in Figure 5. Only this matrix is included, rather than one matrix for every representation experiment. The model obtains perfect performance for *acoustic guitar* (F1 1.000) and strong results for *drum set* (F1 0.979), *electric bass* (F1 0.923), and *piano* (F1 0.898). Auxiliary percussion also improves substantially, reaching an F1 of 0.852.

The main remaining weakness is the separation of the two electric-guitar categories. Only four of the 24 distorted-electric-guitar segments are classified correctly, while 14 are assigned to clean electric guitar and five to synthesizer. As a result, distorted electric guitar has perfect precision but very low recall, producing an F1 of 0.286. Conversely, clean electric guitar reaches a recall of 0.958 but lower precision because it absorbs many distorted-guitar predictions.

The other difficult classes are *FX/processed sound* (F1 0.500) and *synthesizer* (F1 0.524). Five synthesizer segments are classified as FX/processed sound, while FX examples are distributed across auxiliary percussion, vocalists, electric bass, male singer, and synthesizer. Some overlap also remains between the vocal categories, most notably six female-singer segments classified as vocalists.

Overall, Notebook 2 demonstrates that representation choice has a greater effect than the classifier-head variations explored in Notebook 1. Intermediate layers, learned layer aggregation, and the larger MERT-v1-330M checkpoint all improve over the original final-layer MERT-v1-95M representation. The remaining errors are concentrated primarily in acoustically overlapping or heterogeneous categories, especially the two electric-guitar classes and *FX/processed sound*.

7.3 Notebook 3: Direct MERT and Last-Layer Fine-Tuning

Notebook 3 evaluates whether partial adaptation of MERT improves instrument classification relative to a direct frozen-backbone model. In both experiments, the waveform is processed by MERT during every training iteration rather than being replaced by precomputed embeddings. This direct-audio workflow is required for fine-tuning because the selected encoder parameters must remain part of the training graph. The experiments use the same dataset as the previous notebooks.

Both configurations use MERT-v1-95M, the final hidden layer, temporal mean pooling, and an MLP classifier with 256 hidden units and dropout 0.5. Training uses batch size 1, gradient accumulation over eight batches, 16-bit mixed precision, gradient checkpointing, and a maximum of 20 epochs. The classifier head learning rate is 10^{-3} , while trainable backbone parameters use 10^{-5} . The only difference between the two Notebook 3 configurations is the encoder adaptation strategy:

- **Direct frozen MERT:** all MERT parameters remain frozen and only the classifier head is trained;
- **Last-layer fine-tuning:** the final MERT Transformer layer is unfrozen and updated jointly with the classifier head.

7.3.1 Overall results

For context, Table 7 also reports the frozen-MERT reference system from the main pipeline. This reference uses precomputed MERT embeddings and is therefore not a direct-audio fine-tuning

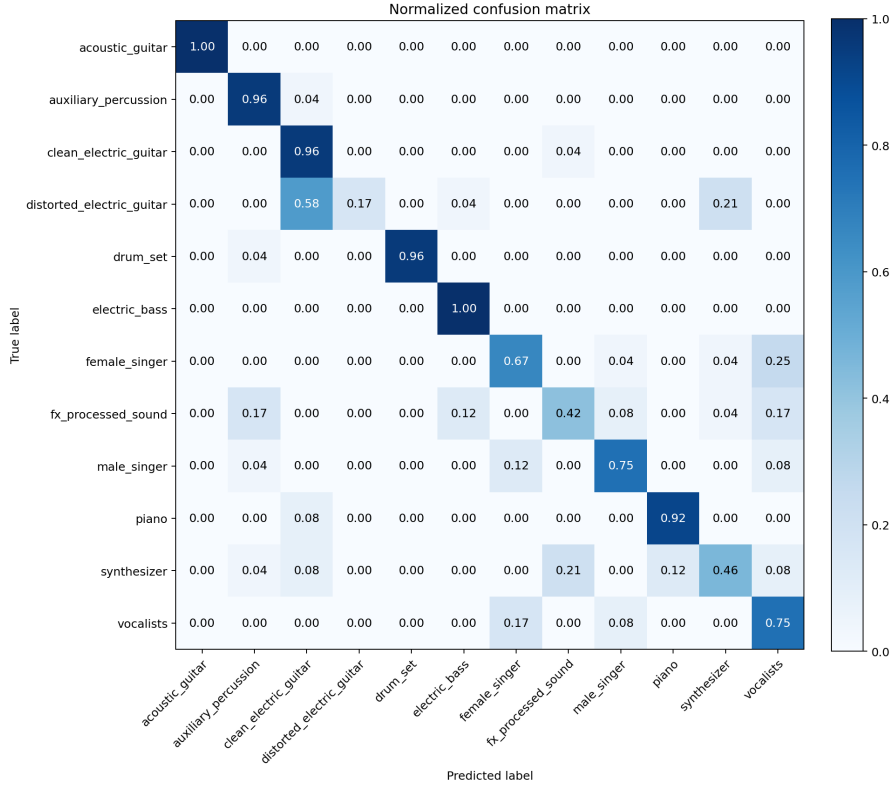


Figure 5: Row-normalised confusion matrix of the MERT-v1-330M model using learned all-layer aggregation and temporal mean pooling.

experiment. It is included only as the comparison point used to judge whether recomputing MERT during training, with or without partial fine-tuning, brings any benefit.

Table 7: Comparison of cached frozen embeddings, direct frozen MERT, and last-layer fine-tuning.

Configuration	Val. macro-F1	Test accuracy	Test macro-F1	Best epoch
Cached frozen embeddings	0.6146	0.6458	0.6431	21
Direct frozen MERT	0.6043	0.6319	0.6258	6
Final Transformer layer unfrozen	0.6032	0.6389	0.6336	6

Relative to the direct frozen configuration, unfreezing the final Transformer layer increases test accuracy from 0.6319 to 0.6389 and test macro-F1 from 0.6258 to 0.6336. This corresponds to two additional correct predictions, increasing the total from 182 to 184 out of 288 test segments.

The gain on the test set is therefore limited to approximately 0.7 percentage points in accuracy and 0.8 percentage points in macro-F1. Moreover, the fine-tuned model obtains a slightly lower validation macro-F1 than the direct frozen model, 0.6032 compared with 0.6043. According to the predefined model-selection protocol, fine-tuning therefore does not produce a confirmed validation improvement.

Both direct-audio configurations also remain below the cached frozen-MERT reference, which reaches a test macro-F1 of 0.6431. The difference cannot be attributed only to caching, since the cached model uses a larger batch size, 32-bit precision, a longer training schedule, and different regularisation settings. Nevertheless, repeatedly executing a frozen encoder provides no performance advantage in the present experiments.

7.3.2 Per-class effects of fine-tuning

Fine-tuning changes the distribution of errors across classes rather than improving every category consistently. The largest F1 variations relative to the direct frozen model are reported in Table 8.

Table 8: Largest per-class F1 changes produced by last-layer fine-tuning.

Class	Direct frozen F1	Fine-tuned F1	Δ F1
Male singer	0.541	0.600	+0.059
FX/processed sound	0.235	0.286	+0.050
Piano	0.844	0.880	+0.036
Synthesizer	0.412	0.369	−0.043
Female singer	0.511	0.476	−0.034
Clean electric guitar	0.577	0.545	−0.031

The largest improvement concerns *male singer*, whose recall increases from 0.417 to 0.500 and whose F1 increases from 0.541 to 0.600. Smaller gains are observed for *FX/processed sound* and *piano*. Piano recall increases from 0.792 to 0.917, producing an F1 of 0.880.

These gains are offset by lower performance for *synthesizer*, *female singer*, and *clean electric guitar*. In particular, synthesizer recall decreases from 0.583 to 0.500. The effect of fine-tuning is therefore class-dependent and does not resolve the most difficult categories systematically.

After fine-tuning, the strongest classes remain *acoustic guitar* and *drum set*, both with F1 0.980, followed by *piano* with F1 0.880 and *electric bass* with F1 0.851. The weakest classes remain *FX/processed sound* with F1 0.286, *synthesizer* with F1 0.369, and *distorted electric guitar* with F1 0.378.

7.3.3 Confusion-matrix analysis

To avoid duplicating all available figures, only the row-normalised confusion matrices of the two direct-audio configurations are shown in Figure 6.

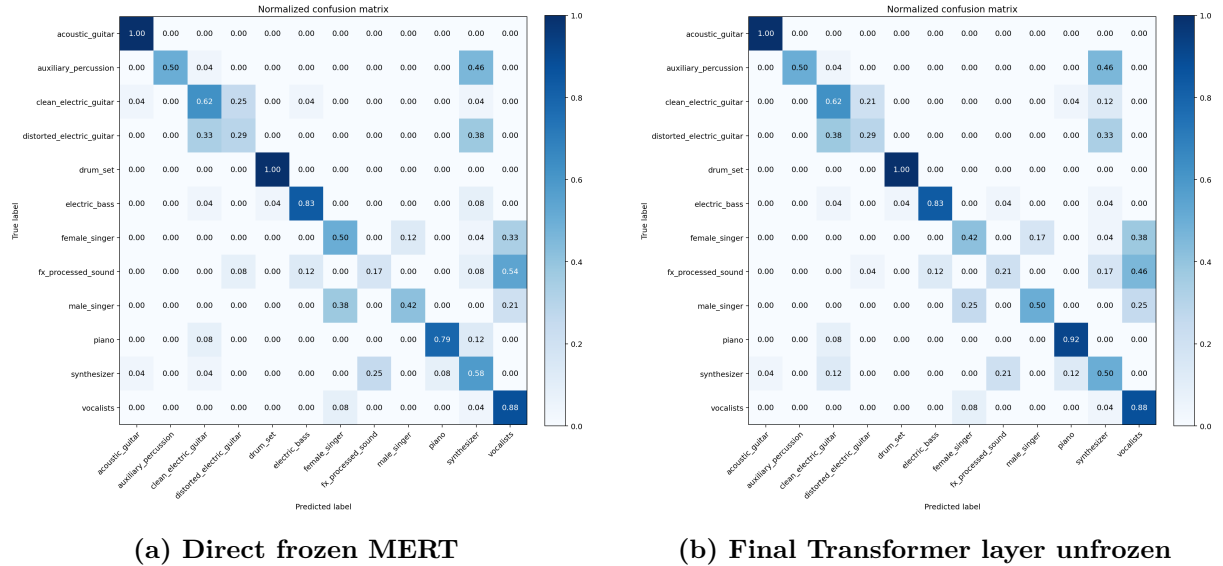


Figure 6: Row-normalised confusion matrices for the direct frozen and last-layer fine-tuned MERT-v1-95M models.

Several dominant confusion patterns remain almost unchanged. In both models, 11 of the 24 *auxiliary percussion* segments are classified as *synthesizer*. Distorted electric guitar is also

consistently confused with clean electric guitar and synthesizer. After fine-tuning, only seven distorted-guitar segments are classified correctly, while nine are assigned to clean electric guitar and eight to synthesizer.

The main changes occur within the vocal classes. In the direct frozen model, nine male-singer segments are classified as female singer and five as vocalists. After fine-tuning, these errors decrease to six examples for each destination, increasing male-singer recall. However, nine female-singer segments are then classified as vocalists, compared with eight in the frozen configuration, and female-singer F1 consequently decreases.

The *FX/processed sound* class remains highly problematic. Fine-tuning increases the number of correctly classified examples from four to five and reduces the number assigned to vocalists from 13 to 11, but most examples are still distributed across unrelated classes. This confirms that the category does not form a stable and homogeneous timbral group.

Overall, last-layer fine-tuning produces a small improvement on the held-out test set but no improvement on the validation criterion used for model selection. It also leaves the principal confusion patterns largely unchanged. Given the additional computation required to propagate gradients through the encoder, these results do not provide strong evidence that fine-tuning the final MERT-v1-95M layer is preferable to using frozen representations under the current data and training conditions.

7.4 Notebook 4: Segment and Training-Protocol Ablations

Notebook 4 evaluates whether the reference isolated-stem system remains stable when the input segmentation and data protocol are changed. The model architecture is kept fixed as frozen MERT-v1-95M with final-layer mean pooling and an MLP classifier with 256 hidden units and dropout 0.5. Consequently, the experiments isolate changes in the data setup rather than introducing a new representation or classifier.

The analysis considers five protocol variants. The 5-second balanced run is used as the control condition. It is compared with 2-second and 10-second segment variants, a reduced balanced subset with 40 segments per class, and a naturally imbalanced subset trained without explicit class reweighting. All experiments are evaluated with the same single-label protocol used in the previous isolated-stem sections.

7.4.1 Completed experiments

The held-out results are reported in Table 9. The 5-second balanced run provides the control condition. Weighted-F1 is included because it differs from macro-F1 when the test set is not class-balanced.

Table 9: Evaluated segment and data-protocol variants in Notebook 4.

Configuration	Test ex- amples	Test accuracy	Test macro-F1	Test weighted- F1
5 s balanced control	288	0.6424	0.6398	0.6398
2 s segments	288	0.6424	0.6261	0.6261
10 s segments	288	0.6215	0.6043	0.6043
40 segments per class	72	0.5694	0.5514	0.5514
Natural-frequency subset, plain training	240	0.7167	0.6775	0.6992

Reducing the segment duration from 5 to 2 seconds leaves accuracy unchanged, but decreases macro-F1 from 0.6398 to 0.6261. This suggests that shorter segments still contain enough

information for a similar number of correct predictions, but the errors become less evenly distributed across classes. In particular, male singer and electric bass improve, while piano, FX/processed sound, and vocalists decrease.

The 10-second experiment obtains the lowest result among the full-size balanced subsets, with macro-F1 0.6043. However, this run is not a perfectly controlled duration comparison: the subset builder selects *drum machine* instead of *auxiliary percussion* because the availability of complete 10-second segments changes the feasible class set. Its score therefore reflects both the longer temporal context and a slightly different classification task.

The reduced subset retains the original 12 classes but uses only 40 segments per class, corresponding to 480 examples overall and six test segments per class. Its macro-F1 decreases to 0.5514, approximately 8.8 percentage points below the 5-second control. The frozen representation therefore remains informative with limited data, but the classifier becomes substantially less reliable as the number of training examples decreases. Per-class results must also be interpreted cautiously because of the small test support.

7.4.2 Natural-frequency class distribution

The natural-frequency subset keeps unequal class counts while still ensuring that every class is represented. Unlike the balanced control, its 240-example test set has class supports ranging from nine male-singer segments to 39 drum-set segments. Training uses ordinary shuffled sampling and unweighted cross-entropy, so no explicit correction is applied to the class imbalance.

This configuration reaches an accuracy of 0.7167, a macro-F1 of 0.6775, and a weighted-F1 of 0.6992, corresponding to 172 correct predictions. The difference between macro-F1 and weighted-F1 reflects the non-uniform class support and the particularly strong performance of some frequent classes, including drum set and electric bass.

Although these values exceed those of the balanced control, they should not be interpreted as evidence that natural class imbalance improves classification. This protocol changes both the selected segments and their class distribution, and therefore represents a different evaluation task. The result instead shows that the system remains effective when the class frequencies are not artificially equalised.

7.4.3 Effect on class confusions

The row-normalised matrices for the two segment-duration variants are shown in Figure 7.

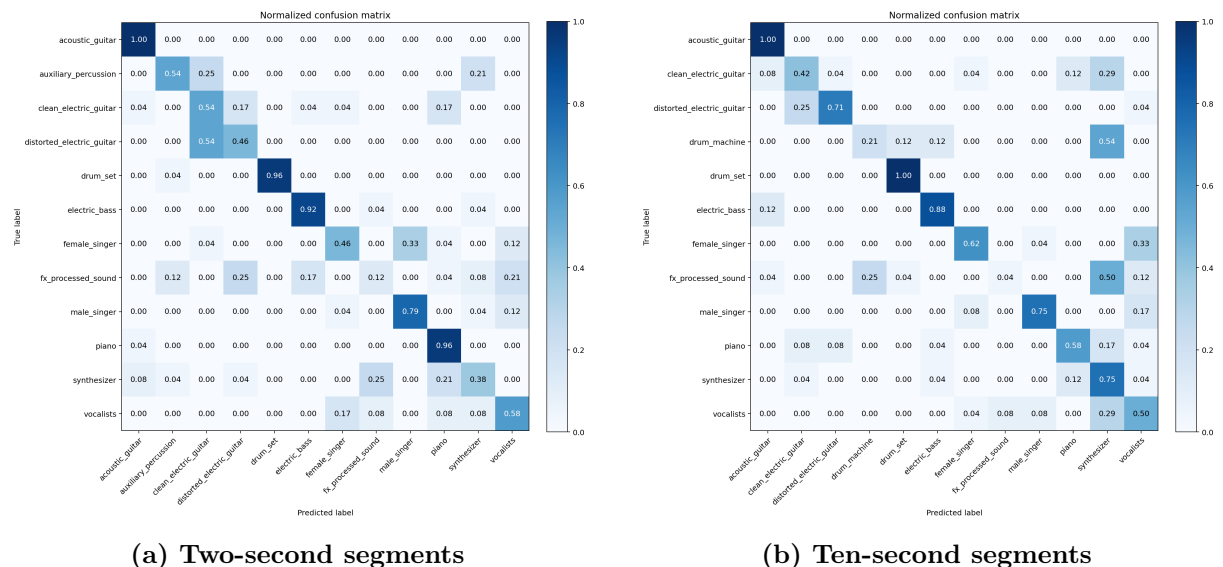


Figure 7: Row-normalised confusion matrices for the segment-duration ablations in Notebook 4. The 10-second subset contains *drum machine* instead of *auxiliary percussion*.

With 2-second segments, the dominant error is the assignment of 13 distorted electric-guitar examples to clean electric guitar. Confusion also remains between the vocal categories, while FX/processed sound reaches an F1 of only 0.167. The shorter context therefore does not eliminate the timbral ambiguities already observed in the reference system.

For 10-second segments, synthesizer becomes the main destination for ambiguous predictions. Thirteen drum-machine examples and 12 FX/processed-sound examples are classified as synthesizer. FX/processed sound reaches a recall of only 0.042, whereas distorted electric guitar, female singer, and male singer improve relative to the 5-second control. Longer segments therefore benefit some sustained sources but do not provide a uniform improvement across classes.

The natural-frequency subset is reported separately in Figure 8, since its rows have different test supports.

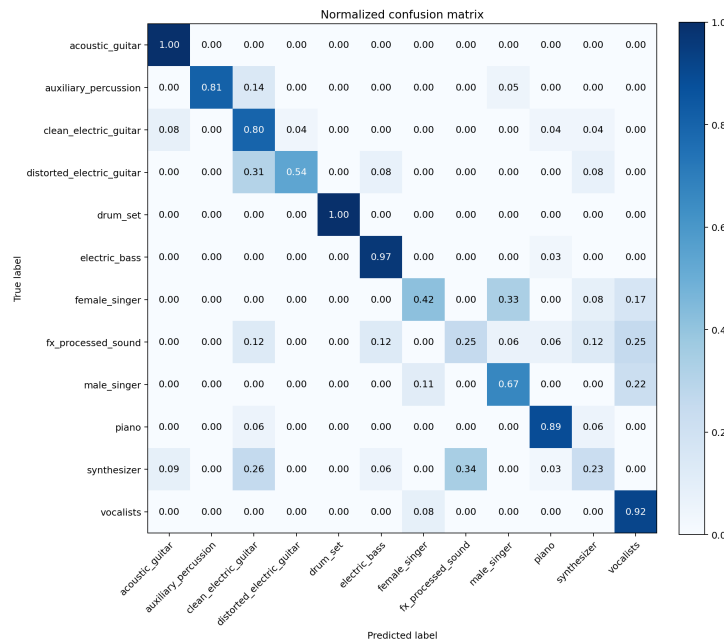


Figure 8: Row-normalised confusion matrix for the natural-frequency subset with plain, un-weighted training.

The natural-frequency model recognises all 39 drum-set examples and 28 of the 29 electric-bass examples. It also performs strongly for auxiliary percussion, piano, acoustic guitar, and vocalists. The main weakness is synthesizer: only eight of 35 examples are classified correctly, while 12 are assigned to FX/processed sound and nine to clean electric guitar. FX/processed sound also remains difficult, with only four of 16 examples correctly recognised, while female singer reaches a recall of 0.417.

Overall, the evaluated variants indicate that the 5-second protocol offers the best compromise among the balanced segment-duration conditions. Reducing the available training data causes the clearest performance loss, while changing segment duration mainly redistributes errors between classes. The natural-frequency run shows that the model remains effective when class counts are not artificially equalised, although it should be interpreted as a separate data condition rather than as a direct improvement over the balanced control.

Isolated-Stem Model Selection and Polyphonic Handoff

This stage closes the isolated-stem part of the evaluation and defines the reference configuration used in the following polyphonic experiments. In the beginning of this chapter, the complete pipeline run was introduced as the frozen-MERT reference system. The same terminology is used here: the model transferred to the polyphonic stage is the reference configuration, not a new model selected specifically for the polyphonic task.

The isolated-stem experiments reported in the previous notebooks show that larger MERT-v1-330M configurations achieved stronger isolated-stem performance. In particular, the MERT-v1-330M model with final-layer mean pooling obtained the highest validation macro-F1, while the MERT-v1-330M model with learned all-layer aggregation obtained the strongest held-out test macro-F1. However, the completed polyphonic experiments use the smaller MERT-v1-95M final-layer mean-pooling configuration introduced by the reference pipeline.

This choice is intentional. The aim of the polyphonic stage is not to repeat isolated-stem model selection, but to study how the reference pipeline behaves when the task is extended from single-label isolated stems to multi-label synthetic mixtures. Keeping the original reference configuration makes the polyphonic results easier to compare with the main isolated-stem pipeline and keeps the computational cost of mixture-level experiments lower.

Table 10: Isolated-stem configurations relevant to the polyphonic reference setup.

Configuration	Val. macro-F1	Test acc.	Test macro-F1	Role
MERT-v1-95M, final layer, mean pooling	0.6146	0.6458	0.6431	Polyphonic reference
MERT-v1-330M, final layer, mean pooling	0.7337	0.7222	0.7091	Best validation model
MERT-v1-330M, learned all-layer aggregation	0.7002	0.7500	0.7290	Best held-out test model

The reference configuration consists of a frozen MERT-v1-95M encoder, the final Transformer layer, temporal mean pooling, a 768-dimensional representation, and an MLP classifier with 256 hidden units and dropout 0.2. In the isolated-stem pipeline this classifier predicts one instrument class with a softmax output. In the polyphonic stage, the same representation backbone is retained, but the output layer is replaced by 12 independent sigmoid outputs trained with a multi-label binary cross-entropy objective. This modification makes the system compatible with mixtures containing more than one active instrument class.

7.5 Notebook 5: Controlled Polyphonic Mixture Dataset

Notebook 5 introduces the transition from single-label isolated-stem classification to multi-label instrument recognition. Its purpose is not to train a classifier, but to construct the controlled polyphonic dataset used in the following experiments.

In this study, the polyphonic stage is based on synthetic mixtures created from the same 12-class MedleyDB subset used in the isolated-stem experiments. Each mixture is built by combining independently selected five-second stem segments, while explicitly controlling the number of active instrument classes. This allows the effect of instrumental overlap to be studied separately from the additional variability introduced by real arrangements, production choices, and mixing decisions.

Only this controlled synthetic-mixture setting is considered in the reported experiments. Therefore, all results in this subsection, and in the following polyphonic evaluations, refer to this

generated mixture dataset.

7.5.1 Controlled synthetic-mixture protocol

The source pool is the same 12-class MedleyDB subset introduced in the isolated-stem part of the evaluation. The original track-level train, validation, and test assignment is preserved before mixture construction, so all sources used in a generated mixture belong to the same data split. This keeps the polyphonic dataset consistent with the track-disjoint evaluation protocol used earlier.

For each split, mixtures are generated with four different overlap levels: one, two, three, or four active instrument classes. For a selected overlap level, the generation procedure first samples the required number of distinct instrument labels and then selects one five-second source segment for each of them. Since the sources are sampled independently, a generated mixture may combine stems from different songs. This design is intentional: the aim is to obtain a controlled benchmark where the number of simultaneously active instruments is known exactly.

Each generated example is assigned a multi-label target vector over the 12 instrument classes. A class is marked as active when one of its stems is present in the mixture and inactive otherwise. Therefore, the number of active labels in the target vector is equal to the intended overlap level of the mixture:

$$k_{\text{active}} = \sum_{c=1}^{12} y_{m,c}. \quad (2)$$

During mixture construction, each source waveform is peak-normalised before combination. The selected sources are then averaged and the resulting mixture is peak-normalised again. This avoids clipping and limits the influence of large level differences between the original recordings. At the same time, this procedure should be interpreted as a controlled simulation of overlap rather than as a realistic model of a professionally mixed song.

Table 11: Configuration of the controlled synthetic-mixture dataset generated in Notebook 5.

Component	Configuration
Source material	12-class MedleyDB isolated-stem subset used in the previous experiments
Mixture type	Synthetic mixtures of independently sampled stem segments
Instrument vocabulary	12 classes
Segment duration	5 s
Overlap levels	$k_{\text{active}} \in \{1, 2, 3, 4\}$
Mixtures per split and overlap level	64
Data split	Original track-disjoint train/validation/test assignment
Random seed	42
Audio representation	Mixtures reconstructed from their source segments when needed

7.5.2 Generated manifest and validation

The generation stage produced a controlled synthetic-mixture manifest with 768 examples. Each of the train, validation, and test partitions contains 64 mixtures for each value of $k_{\text{active}} \in \{1, 2, 3, 4\}$,

giving 256 examples per split. Its nominal composition is therefore.

$$3 \text{ splits} \times 4 \text{ overlap levels} \times 64 \text{ mixtures} = 768 \text{ examples.} \quad (3)$$

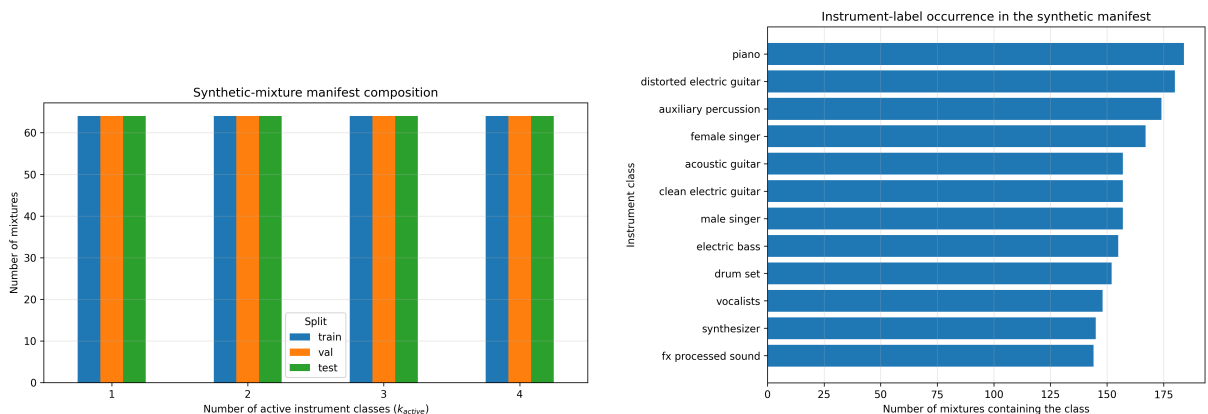
The exact distribution is reported in Table 12.

Table 12: Number of synthetic mixtures for each split and overlap level.

Split	$k = 1$	$k = 2$	$k = 3$	$k = 4$	Total
Training	64	64	64	64	256
Validation	64	64	64	64	256
Test	64	64	64	64	256
Total	192	192	192	192	768

Across the complete dataset there are 1920 active-label occurrences. Perfect class balancing would yield 160 occurrences per class. The observed counts range from 144 for *FX/processed sound* to 184 for *piano*. Consequently, the generated dataset is exactly balanced with respect to split and overlap level, but only approximately balanced with respect to individual instrument classes.

The generated dataset contains 372 distinct active-label combinations: all 12 single-instrument cases, 65 of the 66 possible two-instrument pairs, 129 three-instrument combinations, and 166 four-instrument combinations. The dataset therefore provides almost complete pairwise coverage, together with a broad range of higher-order overlaps.



(a) Number of examples for each split and overlap level. (b) Instrument-label occurrence across the complete manifest.

Figure 9: Composition of the controlled synthetic-mixture dataset generated in Notebook 5. The dataset is exactly balanced with respect to split and k_{active} , while individual class occurrence exhibits limited variation due to random label sampling.

A final integrity check found 758 distinct mixture identifiers among the 768 manifest rows. Ten rows repeat a previously generated example, all in the $k_{\text{active}} = 1$ condition. These repetitions remain inside their original data partition and therefore do not create train–test leakage. However, they reduce the effective number of unique examples by approximately 1.3%.

The dataset generated in Notebook 5 should be interpreted as a controlled overlap benchmark rather than as a realistic simulation of complete musical mixes. This is an intentional design choice: by sampling the number of active instrument classes explicitly, the experiment can evaluate how the model behaves as polyphony increases from one to four simultaneous sources.

At the same time, the mixtures do not reproduce all properties of real produced music. The sources are sampled independently and combined with a simple normalisation procedure, so the resulting examples do not fully preserve natural instrument co-occurrences, relative source levels, room acoustics, masking relationships, or production effects. Therefore, the following results measure the robustness of the reference model under controlled overlap. They should not be read as a direct estimate of performance on commercial full-mix recordings.

7.6 Notebook 6: Frozen-MERT Multi-Label Classification

Notebook 6 evaluates whether the frozen MERT representation selected for isolated-stem recognition can be transferred to polyphonic instrument identification. Unlike the single-label experiments, a mixture may contain several active classes simultaneously; the classifier must therefore estimate one independent probability for each of the 12 instrument labels rather than selecting a single mutually exclusive class.

The reported experiment uses the controlled synthetic-mixture dataset generated in Notebook 5. Therefore, the results in this subsection evaluate the reference MERT representation under a multi-label setting where the number of active instrument classes is known and controlled.

7.6.1 Representation and classifier

Each five-second mixture is converted to mono, resampled to 24 kHz, and represented with the frozen MERT-v1-95M encoder. As in the reference isolated-stem configuration, the final Transformer layer is averaged over time, producing a 768-dimensional embedding for each mixture. The encoder is kept fixed, so the experiment measures how well a lightweight classifier can use the frozen representation in a multi-label setting.

The classifier is an MLP with one hidden layer of 256 units, ReLU activation, dropout 0.2, and 12 output units. In contrast to the isolated-stem classifier, the outputs are not mutually exclusive. Each output is passed through a sigmoid function and interpreted as the probability that the corresponding instrument is present:

$$p_{m,c} = \sigma(z_{m,c}). \quad (4)$$

The final prediction is obtained with a fixed threshold of 0.5:

$$\hat{y}_{m,c} = \begin{cases} 1, & p_{m,c} \geq 0.5, \\ 0, & p_{m,c} < 0.5. \end{cases} \quad (5)$$

The model is trained with weighted binary cross-entropy. The positive examples of each class are weighted according to their frequency in the training split, so that classes with fewer positive examples are not ignored by the classifier. This formulation is appropriate for polyphonic recognition because the presence of one instrument does not exclude the presence of another.

7.6.2 Overall test performance

The global test results are reported in Table 14. Exact-match accuracy requires the entire predicted label set to coincide with the reference set and is therefore much stricter than class-wise F1.

The model recovers a substantial fraction of the active labels, but it also tends to predict too many instruments. The test mixtures contain an average of 2.50 reference labels, whereas the classifier predicts 3.67 labels per mixture at the fixed threshold of 0.5. This explains why

Table 13: Configuration of the multi-label classifier trained in Notebook 6.

Component	Configuration
Input dataset	Synthetic mixtures generated in Notebook 5
Data split	256 training, 256 validation, and 256 test mixtures
Encoder	Frozen MERT-v1-95M
Representation	Final layer, temporal mean pooling, 768 dimensions
Classifier	MLP 768 $\rightarrow$ 256 $\rightarrow$ 12, ReLU, dropout 0.2
Loss	Weighted binary cross-entropy with class-dependent positive weights
Optimiser	AdamW
Learning rate	10^{-3}
Weight decay	10^{-2}
Batch size	32
Maximum epochs	30
Early stopping	Validation macro-F1, patience 8
Decision threshold	Fixed at 0.5
Random seed	42

Table 14: Held-out test performance of the synthetic-mixture multi-label model.

Metric	Value
Best validation macro-F1	0.4809
Exact-match accuracy	0.0586 (15 / 256)
Micro-precision	0.4473
Micro-recall	0.6563
Micro-F1	0.5320
Macro-precision	0.4850
Macro-recall	0.6478
Macro-F1	0.5353
Weighted-F1	0.5398
Mean average precision	0.5381
Macro ROC-AUC	0.7867

micro-recall is considerably higher than micro-precision, and why exact-match accuracy remains low.

The macro ROC-AUC of 0.7867 shows that the predicted probabilities still contain useful ranking information. In other words, the model often assigns higher scores to relevant instruments, but the fixed threshold converts too many of these scores into positive predictions.

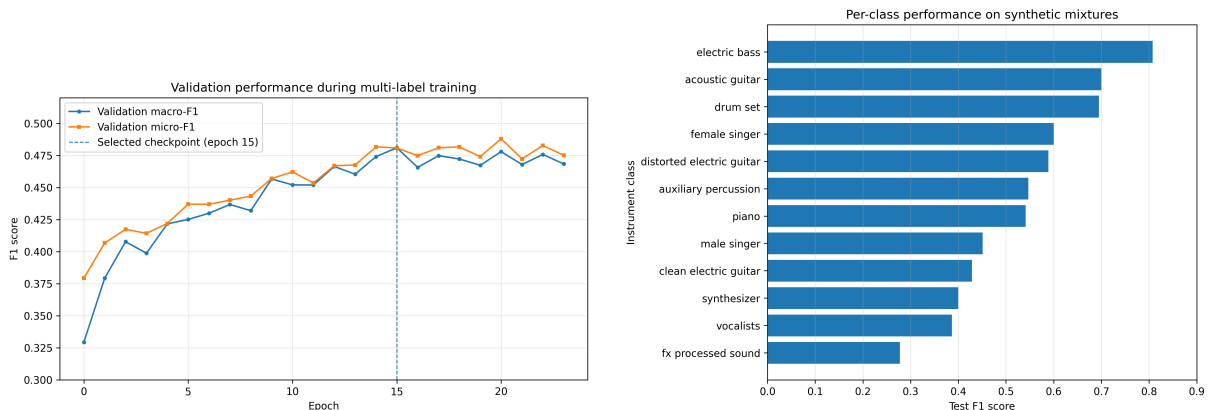
7.6.3 Per-class results

Table 15: Per-class test performance for the synthetic-mixture classifier.

Class	Support	Precision	Recall	F1
Acoustic guitar	51	0.609	0.824	0.700
Auxiliary percussion	57	0.592	0.509	0.547
Clean electric guitar	58	0.304	0.724	0.429
Distorted electric guitar	69	0.511	0.696	0.589
Drum set	46	0.673	0.717	0.695
Electric bass	49	0.764	0.857	0.808
Female singer	63	0.582	0.619	0.600
FX/processed sound	46	0.385	0.217	0.278
Male singer	48	0.426	0.479	0.451
Piano	62	0.386	0.903	0.541
Synthesizer	46	0.277	0.717	0.400
Vocalists	45	0.311	0.511	0.387

Electric bass is the strongest class, with F1 0.808, followed by acoustic guitar and drum set. These classes also performed well in the isolated-stem experiments, suggesting that their timbral characteristics remain comparatively stable when other sources are added.

The weakest class is FX/processed sound, with F1 0.278 and recall 0.217, followed by vocalists and synthesizer. Several classes also show the over-prediction behaviour observed in the global metrics. Piano reaches very high recall, 0.903, but its precision is only 0.386. Similarly, synthesizer and clean electric guitar obtain recalls above 0.71, but their precisions remain below 0.31. This indicates that the model is sensitive to broad timbral cues, but it sometimes activates labels even when the corresponding instrument is not actually present.



(a) Validation macro- and micro-F1 during training.

(b) Per-class F1 on the held-out test mixtures.

Figure 10: Training and test-set analysis of the frozen-MERT multi-label classifier. The checkpoint selected at epoch 15 maximises validation macro-F1; test performance varies substantially across instrument classes.

Notebook 6 shows that the frozen MERT reference representation can be extended from isolated-stem classification to multi-label recognition on controlled synthetic mixtures. The model retrieves many of the active instruments, but complete label-set prediction remains difficult because the classifier tends to over-predict labels at the fixed threshold.

These results provide the global performance baseline for the polyphonic setting. Notebook 7 therefore examines the same predictions in more detail by grouping the test mixtures according to the number of active instruments.

7.7 Notebook 7: Robustness to Increasing Instrumental Overlap

Notebook 7 investigates whether the multi-label classifier trained in Notebook 6 remains reliable as the number of simultaneously active instrument classes increases. No new model is trained in this stage. Instead, the 256 held-out test predictions are grouped according to the number of active instrument labels, and the evaluation metrics are recomputed separately for each value of k_{active} .

7.7.1 Grouped evaluation procedure

The grouped evaluation uses the same test predictions analysed in Notebook 6. Each mixture is assigned to one group according to the number of active instrument labels in its reference target. Since the controlled test set contains exactly 64 mixtures for each overlap level, the groups are directly comparable in size.

For a mixture m , the number of active instruments is computed as

$$k_{\text{active}} = \sum_{c=1}^{12} y_{m,c}, \quad (6)$$

where $y_{m,c}$ indicates whether instrument class c is present in the mixture. The test set is then divided into four groups, $k_{\text{active}} \in \{1, 2, 3, 4\}$. Within each group, exact-match accuracy, micro-precision, micro-recall, micro-F1, and macro-F1 are recomputed using the 12 binary instrument labels.

Exact-match accuracy is the strictest metric in this analysis. A prediction is counted as correct only when all active instruments are detected and all inactive instruments are rejected. The F1 scores are less strict because they also reward partially correct label sets.

This is especially important for the $k_{\text{active}} = 1$ group. Although these examples contain only one active instrument, they are still evaluated with the multi-label classifier. Therefore, a prediction is counted as an exact match only if the correct instrument is selected and all other 11 instrument classes are rejected. If the model detects the correct instrument but also activates additional labels, the example is not counted as an exact match.

7.7.2 Results by number of active instruments

Table 16: Test performance grouped by the number of active instrument classes.

k_{active}	N	Exact match	Micro-P	Micro-R	Micro-F1	Macro-F1
1	64	0.1563	0.3261	0.7031	0.4455	0.4784
2	64	0.0625	0.4160	0.7734	0.5410	0.5346
3	64	0.0156	0.4762	0.6250	0.5405	0.5304
4	64	0.0000	0.5016	0.6094	0.5503	0.5400

The strict exact-match criterion shows a clear loss of complete label-set recovery as overlap increases. The classifier predicts the full label set correctly for 10 of 64 single-instrument mixtures, four of 64 two-instrument mixtures, one of 64 three-instrument mixtures, and none of the four-instrument mixtures. Thus, the ability to recover every active class while avoiding every false positive decreases monotonically with k_{active} .

The label-wise metrics show a different pattern. Micro- and macro-F1 do not decrease monotonically: micro-F1 rises from 0.4455 at $k = 1$ to approximately 0.54–0.55 for $k = 2, 3, 4$. At the same time, micro-precision increases from 0.3261 to 0.5016, whereas recall peaks at $k = 2$ and then falls from 0.7734 to 0.6094 by $k = 4$.

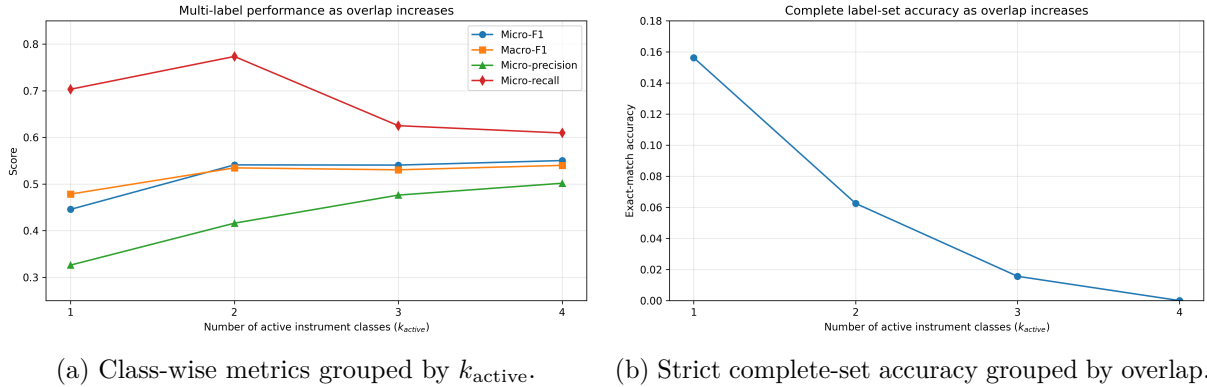


Figure 11: Robustness analysis of the frozen-MERT multi-label classifier. Exact match deteriorates as the number of active labels grows, while micro- and macro-F1 remain approximately stable for $k \geq 2$.

7.7.3 Interpretation

The apparent stability of F1 should not be interpreted as evidence that instrumental overlap has no effect. At the fixed threshold of 0.5, the model systematically predicts more labels than are present. The mean predicted cardinality is 2.16, 3.72, 3.94, and 4.86 labels for reference cardinalities 1, 2, 3, and 4, respectively. When only one instrument is active, these broad predictions create many false positives and produce low precision. As the true label set becomes larger, a greater fraction of the same broad predictions coincides with positive targets, which increases micro-precision and can keep F1 stable even though exact reconstruction becomes harder.

The results therefore reveal two complementary aspects of robustness. The model retains partial class-level detection ability under denser mixtures, but its ability to return the complete and only correct set of instruments rapidly collapses. This distinction is important because F1 rewards partially correct sets, whereas exact match requires all active instruments to be detected and all inactive instruments to be rejected.

These conclusions should be interpreted within the controlled synthetic mixture setting introduced in Notebook 5. Each group contains 64 examples, and the exact instrument combinations are not identical across overlap levels. Therefore, small variations in macro-F1 may partly reflect which class combinations appear in each group.

The main trend is nevertheless clear: the model keeps partial label-detection ability as overlap increases, but complete label-set reconstruction becomes rapidly harder. This supports the use of exact-match accuracy alongside F1 scores when evaluating polyphonic instrument recognition, because the two metrics highlight different aspects of the model behaviour.

7.8 Notebook 8: Genre, Instrumentation, and Co-Occurrence Analysis

Notebook 8 provides a descriptive analysis of the datasets used in the previous experiments. No new classifier is trained in this stage. The aim is to check whether the isolated-stem and synthetic-mixture datasets contain relevant contextual patterns that may affect the interpretation of the classification results.

For the synthetic-mixture dataset, the co-occurrence analysis is used only as a dataset check. Since the mixtures are artificially generated, the observed instrument pairs should not be interpreted as natural musical co-occurrences. Instead, they indicate whether the generated benchmark is dominated by a small number of repeated instrument combinations or whether it contains a reasonably varied set of overlaps.

For the isolated-stem subset, the genre analysis is used to examine whether instrument classes are associated with particular musical genres. This is important because the isolated subset is balanced by class count, but not necessarily by genre. Therefore, some per-class results may partly reflect genre-related recording characteristics in addition to the acoustic properties of the target instrument.

7.8.1 Instrument co-occurrence

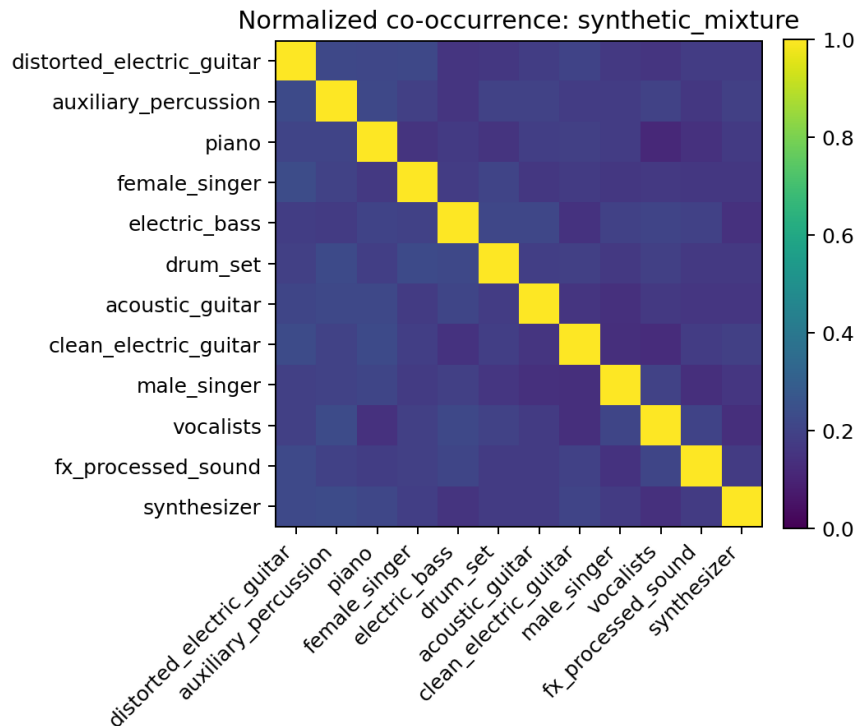


Figure 12: Row-normalised instrument co-occurrence matrix for the synthetic mixture dataset.

The co-occurrence analysis is meaningful for the synthetic-mixture dataset because each example may contain more than one active instrument. The matrix in Figure 12 is row-normalised: for each instrument on the vertical axis, it shows the proportion of its mixtures in which another instrument is also active.

The co-occurrence matrix for the synthetic-mixture dataset should be interpreted as a validation of the generated benchmark rather than as an analysis of natural instrument co-occurrence. The mixtures are constructed artificially, so the instrument pairs are a consequence of the sampling procedure used to build the dataset.

The useful question is therefore not which instruments naturally occur together in MedleyDB, but whether the synthetic benchmark is dominated by a few instrument pairs. The observed off-diagonal values are relatively uniform, ranging from approximately 0.11 to 0.23, which indicates that no single pair of instruments dominates the generated mixtures. This supports the use of the dataset as a controlled benchmark for overlap robustness.

7.8.2 Instrument frequency and genre associations

The genre analysis is most informative for the isolated-stem subset, because each segment comes from a real MedleyDB recording with genre metadata. Although the subset contains the same number of segments for each instrument class, the genres associated with those segments are not uniformly distributed.

This means that class balance does not automatically imply context balance. A classifier may benefit from genre-related cues, recording style, or production characteristics that are correlated with certain instruments.

The genre metadata of the isolated-stem subset reveals several strong segment-level associations. Distorted electric guitar is concentrated mainly in Rock recordings, while acoustic guitar and female singer occur frequently in Singer/Songwriter tracks. FX/processed sound is most strongly associated with Electronic/Fusion. Piano is more evenly distributed across several genres, including Singer/Songwriter, Pop, Jazz, and Classical recordings.

Overall, Notebook 8 should be interpreted as a dataset-context analysis rather than as an additional classification experiment. The co-occurrence analysis of the synthetic mixtures confirms that the generated benchmark is not dominated by a small number of repeated instrument pairs, but it should not be read as an analysis of natural musical co-occurrence.

The genre analysis is more relevant for the isolated-stem subset. It shows that some instrument classes are strongly associated with particular genres, even though the number of segments per class is balanced. These associations may partly influence per-class performance and confusion patterns, because the model can exploit genre-related acoustic or production cues in addition to instrument timbre.

Table 17: Strongest genre associations for selected isolated-stem classes.

Instrument	Dominant genre	Segments	Share
Acoustic guitar	Singer/Songwriter	118 / 160	73.8%
Distorted electric guitar	Rock	134 / 160	83.8%
FX/processed sound	Electronic/Fusion	89 / 160	55.6%
Female singer	Singer/Songwriter	81 / 160	50.6%
Male singer	Rock	65 / 160	40.6%

7.9 Notebook 9: Multi-Label Error Analysis and Representative Examples

Notebook 9 performs a qualitative error analysis of the polyphonic model evaluated in the previous sections. No additional model is trained in this stage. Instead, the goal is to examine the saved test predictions in more detail and identify the most common failure patterns.

For each test mixture, the predicted multi-label vector is compared with the reference set of active instruments. The analysis separates exact-match errors from correct exact matches, counts false positives and false negatives for each instrument class, identifies the weakest classes, and selects a small set of representative cases for qualitative inspection.

7.9.1 Overall error profile

The model is evaluated on 256 synthetic test mixtures using a decision threshold of 0.5. Its main results are summarised in Table 18.

Table 18: Aggregate multi-label results analysed in Notebook 9.

Metric	Value
Exact-match accuracy	0.0625
Micro-F1	0.5313
Macro-F1	0.5349
Micro-precision	0.4463
Micro-recall	0.6563
Macro ROC-AUC	0.7867
Mean average precision	0.5382
Correct exact matches	16 / 256
Mixtures with at least one error	240 / 256

Exact-match accuracy is particularly strict in the multi-label setting because a mixture is considered correct only when the complete predicted label set matches the reference set. Therefore, a prediction is counted as wrong both when an active instrument is missed and when an additional inactive instrument is predicted. The low exact-match score can therefore coexist with micro- and macro-F1 values around 0.53: the model often identifies part of the active instrument set, but does not reliably recover the complete set.

The number of exact matches decreases with mixture cardinality: 10 of the 64 single-instrument examples are fully correct, compared with five for $k_{\text{active}} = 2$, one for $k_{\text{active}} = 3$, and none for $k_{\text{active}} = 4$. This confirms the same qualitative trend observed in Notebook 7: complete label-set recovery becomes progressively harder as the number of active instruments increases.

7.9.2 False positives, false negatives, and weak classes

The most frequent class-level errors are summarised in Table 19.

Table 19: Most frequent false positives and false negatives on the polyphonic test set.

False-positive class	Count	False-negative class	Count
Clean electric guitar	96	FX/processed sound	36
Piano	90	Auxiliary percussion	28
Synthesizer	87	Male singer	25
Vocalists	51	Female singer	24
Distorted electric guitar	47	Distorted electric guitar	21

The dominant error pattern is over-prediction. Clean electric guitar, piano, and synthesizer are the most frequent false-positive classes. Piano reaches a very high recall of 0.903, but its 90 false positives reduce precision to 0.384. A similar pattern appears for clean electric guitar and synthesizer: both classes are detected often, but they are also activated in many mixtures where they are not present.

The most difficult class is *FX/processed sound*, with recall 0.217 and F1 0.282. This class is acoustically heterogeneous, so its examples do not necessarily share one stable timbral pattern. Male singer also has low recall, 0.479, while auxiliary percussion and female singer are among the most frequently missed classes. These results suggest that categories with less consistent

timbral structure, or categories that overlap strongly with other sources, are more difficult to recover in the polyphonic setting.

Electric bass is the strongest class, with F1 0.808, followed by acoustic guitar and drum set. However, the error analysis confirms that high recall does not necessarily imply reliable set prediction when a class is also systematically over-predicted.

7.9.3 Representative-example selection

The qualitative part of the analysis selects 40 representative cases. The selection includes one high-confidence false positive and one strong false negative for each of the 12 instrument classes, together with eight high-confidence exact-match errors and eight high-confidence correct examples.

The purpose of this selection is not to provide another quantitative metric, but to make the error patterns easier to inspect. False-positive examples show which absent instruments are nevertheless assigned high probability. False-negative examples show which active instruments receive low probability. The high-confidence exact-match errors are useful because they reveal cases in which the model is confident but the complete label set is still wrong. The correct examples provide a small reference set of cases where the same model does recover the full label set.

Although 40 cases are selected, they should be understood as representative error-analysis cases rather than as 40 independent conclusions. Their role is to support the interpretation of the aggregate metrics and class-level error counts.

Overall, Notebook 9 shows that the polyphonic model does learn useful instrument evidence, but it does not reliably estimate the complete active label set. The main failure mode is not a total absence of recognition. Rather, the model often detects some relevant sources while adding extra labels or missing less stable classes. This explains why the F1 scores remain moderate, whereas exact-match accuracy is very low.

8 Discussion

The experiments show that automatic instrument classification on MedleyDB is strongly affected by the chosen audio representation. The classical MFCC/spectral SVM baseline is competitive and reaches a test macro-F1 of 0.6752, confirming that handcrafted timbral descriptors remain a meaningful reference for isolated-stem instrument recognition. The initial frozen MERT-v1-95M final-layer system performs slightly worse, with a test macro-F1 of 0.6431, showing that using a self-supervised model does not automatically guarantee better performance when the representation layer and pooling strategy are fixed.

The representation ablations provide the most important result of the isolated-stem study. Intermediate MERT layers, learned layer aggregation, and the larger MERT-v1-330M checkpoint improve substantially over the initial final-layer MERT-v1-95M reference. The best validation-selected configuration is MERT-v1-330M with final-layer mean pooling, while the highest observed held-out score is obtained by MERT-v1-330M with learned all-layer aggregation, reaching a test macro-F1 of 0.7290. This suggests that MERT contains useful information for instrument recognition, but that timbre-related cues are not necessarily concentrated in the final Transformer layer. This is consistent with the broader motivation behind self-supervised music models: their representations can be useful across MIR tasks, but the downstream task still determines which hidden states are most informative.

The error analysis indicates that the strongest classes are those with relatively stable and distinctive timbral profiles, such as acoustic guitar, drum set, electric bass, and piano. The

most persistent errors occur for acoustically overlapping or internally heterogeneous labels, especially clean electric guitar versus distorted electric guitar, synthesizer, vocal categories, and FX/processed sound. These confusions are musically plausible: distortion changes the spectral envelope of electric guitar, synthesizers can imitate many timbres, and processed sounds do not correspond to a single acoustic source type. Therefore, the remaining errors are not only failures of the classifier, but also reflect the difficulty of defining clean instrument classes in real music-production contexts.

Partial fine-tuning of the final MERT-v1-95M Transformer layer gives only a small test improvement over the direct frozen model and does not improve the validation macro-F1 used for model selection. Under the present dataset size and computational constraints, frozen representations are therefore more reliable than limited fine-tuning. This does not mean that fine-tuning is ineffective in general; rather, it suggests that successful adaptation would likely require more data, stronger regularization, more extensive hyperparameter tuning, or a different fine-tuning strategy.

The polyphonic experiments show a clear difference between partial label detection and complete label-set recovery. The multi-label MERT classifier obtains moderate macro-F1 and a macro ROC-AUC of 0.7867, indicating that the predicted probabilities contain useful information about active instruments. However, exact-match accuracy is only 0.0625, and it decreases as the number of overlapping instruments increases. In the four-instrument condition, no test mixture is predicted with a fully correct label set. This happens mainly because the model tends to over-predict labels: it often detects some active sources, but also activates additional absent instruments.

This result is important for interpreting robustness. Macro-F1 and micro-F1 remain relatively stable when moving from two to four active instruments, but exact-match accuracy deteriorates. The model therefore preserves some instrument-level evidence under overlap, but does not reliably identify the complete set of active instruments. For polyphonic instrument recognition, exact-match accuracy and F1 should be reported together, because they describe different behaviours: F1 rewards partially correct predictions, while exact match measures whether the full active instrumentation is recovered.

The dataset-context analysis also shows that class balance does not imply complete experimental neutrality. Some classes are associated with specific genres or production styles, such as distorted electric guitar with Rock and FX/processed sound with Electronic/Fusion. Consequently, models may exploit contextual or production-related cues in addition to instrument timbre. This does not invalidate the results, but it should be considered when interpreting generalization beyond the selected MedleyDB subset.

9 Limitations

The first limitation is the size and composition of the dataset. Although MedleyDB is well suited to this project because it provides multitrack recordings and stem-level information, the final isolated-stem subset contains only 12 selected instrument classes and 160 segments per class. Rare instruments are excluded, and some retained classes have limited track diversity. For example, very strong performance on a class may partly reflect the small number of distinct recordings available for that class rather than full generalization across many performers, genres, and recording conditions.

Second, the isolated-stem task is cleaner than real-world music listening. Stems marked with bleed are excluded, low-energy segments are removed, and each isolated segment is assigned a single label. This design is appropriate for establishing a controlled baseline, but it simplifies the ambiguity present in real mixtures, where instruments overlap, share frequency ranges, and may

be affected by mixing effects.

Third, the polyphonic dataset is synthetic. Mixtures are created by averaging isolated stem segments with known labels, which makes the benchmark controlled and reproducible. However, these mixtures do not fully reproduce professional mixing decisions, natural loudness relations, spatial placement, masking, arrangement structure, or genre-dependent co-occurrence. Therefore, the polyphonic results should be interpreted as a robustness test under controlled overlap, not as a complete evaluation on natural full-song mixtures.

Fourth, the polyphonic experiments use the MERT-v1-95M final-layer reference representation rather than the best isolated-stem MERT-v1-330M configuration. This choice keeps the polyphonic stage comparable with the main reference pipeline and reduces computational cost, but it also means that the reported multi-label results are not necessarily the best possible MERT-based mixture results.

Finally, the evaluation is based on a limited number of splits and selected hyperparameter settings. Validation-based model selection is used consistently, but repeated random seeds, statistical significance tests, threshold calibration, and broader fine-tuning strategies are not explored. In particular, the fixed multi-label threshold of 0.5 contributes to over-prediction, suggesting that class-specific or validation-optimized thresholds could improve exact-match performance.

10 Conclusion and Future Work

This project implemented and evaluated a complete instrument-classification pipeline on MedleyDB v1, covering isolated-stem classification, representation ablations, partial fine-tuning, protocol variants, synthetic mixture generation, multi-label classification, and robustness analysis under increasing instrumental overlap. The work addresses both parts of the task by comparing a classical feature-engineering baseline with self-supervised music representations and by extending the evaluation from isolated stems to polyphonic mixtures.

The main conclusion is that MERT-based transfer learning is effective for instrument recognition when the representation is carefully selected. The initial frozen MERT-v1-95M final-layer system does not outperform the MFCC/spectral SVM baseline, but intermediate layers, learned layer aggregation, and the larger MERT-v1-330M checkpoint improve the results substantially. The highest observed isolated-stem performance is obtained by MERT-v1-330M with learned all-layer aggregation, reaching a test macro-F1 of 0.7290.

At the same time, the results show that polyphonic recognition remains more challenging than isolated-stem classification. The multi-label model often detects part of the active instrumentation, but complete label-set prediction is unreliable, especially as the number of simultaneous instruments increases. This highlights the need for evaluation metrics that distinguish partial recognition from complete mixture understanding.

Overall, the study shows that self-supervised music representations provide a strong direction for MedleyDB instrument classification, but also that representation choice, dataset design, label ambiguity, and polyphonic overlap strongly influence the final performance. Future work should evaluate the best MERT-v1-330M representations in the polyphonic setting, apply threshold calibration, test more realistic mixture-generation strategies, and extend the evaluation to more classes, larger datasets, and natural full-song mixtures.

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